

Block Schur–Secular Contraction Budgets

Exact Threshold Geometry and a Four-Step Replacement for Pointwise Sub-Quarter Decay

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Abstract

RH-90 certified a rank-one packet correction with contraction factor below 0.24 at one selected update in each of ten frozen channels. RH-91 then used pointwise sub-quarter contraction as a sufficient bootstrap hypothesis. The present paper shows that this pointwise formulation is stronger than the finite data support and replaces it by a block budget.

For an enriched Hermitian compression

$$H = \begin{pmatrix} A & b \\ b^* & d \end{pmatrix}, \quad \Delta = d - \lambda_{\min}(H),$$

and requested gain $\delta \geq 0$, set $M_\delta = A - (d - \delta)I$. The exact threshold dichotomy is

$$\Delta \geq \delta \iff \lambda_{\min}(M_\delta) \leq 0 \quad \text{or} \quad (M_\delta \succ 0 \text{ and } b^* M_\delta^{-1} b \geq \delta).$$

In the coercive branch, every trial vector satisfies the defect identity

$$\Phi_\delta(x) = \delta - b^* M_\delta^{-1} b + (M_\delta x - b)^* M_\delta^{-1} (M_\delta x - b).$$

Thus absolute Schur margins should be normalized by the requested gain, and ill conditioning is separated from the final sign.

The variable-budget theorem allows one-step factors ρ_j to vary and requires only their block product to contract. A 384-bit audit freezes four exact rational factors in each channel. All forty Schur forms are strictly negative, every refreshed packet dominates its rank-one corrected predecessor, and the largest four-step budget is $0.003219969564 < 0.24^4$, with geometric mean 0.23821162. Conversely, seven individual updates have a positive-definite 0.24 threshold matrix; the smallest obstructed contraction is 0.25042832. Hence the block law is genuinely weaker than pointwise sub-quarter decay.

This is an exact finite-dimensional reduction and a frozen four-step certificate, not a repeated all-level block theorem. Reduced packet refresh, the prefix/observability bridge, Stage A, Hilbert–Polya, and the Riemann Hypothesis remain open.

Keywords: Schur complement; secular equation; block contraction; Rayleigh–Ritz; validated numerics; dynamic packet.

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1 Introduction

The normalized-memory packet route replaces unstable tail eigenspaces by captured energy. RH-86 introduced the trace-normalized memory Gramian [7]; RH-89 showed that one complement direction and an $(r + 1)$ -dimensional Ritz solve recover most of the observed reoptimization dividend [9]; and RH-90 removed the full reference packet from a selected one-step certificate [11]. RH-91 then isolated a sufficient all-level hypothesis: after burn-in, every update contracts the memory tail by a fixed factor $\rho < 1$ [8].

The first purpose of this paper is to sharpen the algebra behind that hypothesis. The RH-90 trial inequality is sufficient, but the compressed matrix admits an exact necessary-and-sufficient dichotomy. Either the old block already contains a threshold direction, or a positive-definite resolvent-weighted coupling must cross the requested gain. Completing the square then gives an exact trial-defect identity. These statements identify the natural relative margin and explain why a tiny absolute value of Φ_δ need not signal a fragile mechanism.

The second purpose is structural. Uniform pointwise contraction is often stronger than decay of a nonautonomous product. For positive tails, the quantity that enters a multi-update bootstrap is $\prod_j \rho_j$, not $\max_j \rho_j$. We therefore replace a fixed one-step target by a variable contraction budget and prove a block bootstrap with an explicit within-block prefix constant. The proof is elementary, but the change of quantifier is important: intermittent slow updates are permitted provided the block has enough compensating decay.

The third purpose is a validated route audit. We examine the four updates ending at the archived packet time (with a one-step adjustment at the coarsest scale). The result has both positive and negative parts:

1. all forty variable-budget Schur gates are strict and all ten four-step products lie below 0.24^4 ;
2. seven of the same updates rigorously fail the pointwise 0.24 target for the named rank-one compressed correction;
3. a refreshed leading packet is no worse than the small corrected packet at all forty steps, but producing such a refresh in polylogarithmic dimension remains a separate theorem gate.

The last item prevents a common overinterpretation. The block theorem does not prove that repeatedly applying only one complement correction tracks the optimal packet. It assumes a refresh satisfying an energy dominance inequality. In the Stage-A dependency language, the block Schur law and the reduced-refresh law remain distinct.

2 Exact Schur threshold geometry

Let $G \geq 0$ be a finite-dimensional or trace-class Gram operator, let $V : \mathbb{C}^r \rightarrow \mathcal{K}$ be an isometry, and let $q \perp \text{Ran } V$ be a unit complement direction. Put $Z = [V, q]$ and

$$H = Z^* G Z = \begin{pmatrix} A & b \\ b^* & d \end{pmatrix}.$$

Retaining the leading r eigenvectors of H instead of the old coordinate packet gains exactly

$$\Delta = d - \lambda_{\min}(H) \geq 0. \tag{1}$$

This is the one-dimensional deletion formula used in RH-90. We now resolve the threshold event $\Delta \geq \delta$ exactly.

Theorem 2.1 (Exact Schur threshold dichotomy). *For $\delta \geq 0$, define*

$$M_\delta = A - (d - \delta)I.$$

Then

$$\Delta \geq \delta \iff \lambda_{\min}(M_\delta) \leq 0 \quad \text{or} \quad (M_\delta \succ 0 \text{ and } b^* M_\delta^{-1} b \geq \delta). \tag{2}$$

In particular, a zero eigenvalue of M_δ already certifies the requested gain; no pseudoinverse boundary case is needed.

Proof. Set $t = d - \delta$. By (1), $\Delta \geq \delta$ is equivalent to $\lambda_{\min}(H) \leq t$, hence to

$$\lambda_{\min} \begin{pmatrix} M_\delta & b \\ b^* & \delta \end{pmatrix} \leq 0.$$

If $\lambda_{\min}(M_\delta) \leq 0$, restricting the Rayleigh quotient to vectors $(x, 0)$ proves (2). If $M_\delta \succ 0$, Schur congruence gives

$$\begin{pmatrix} I & -M_\delta^{-1}b \\ 0 & 1 \end{pmatrix}^* \begin{pmatrix} M_\delta & b \\ b^* & \delta \end{pmatrix} \begin{pmatrix} I & -M_\delta^{-1}b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} M_\delta & 0 \\ 0 & \delta - b^*M_\delta^{-1}b \end{pmatrix}.$$

Sylvester inertia therefore proves the second branch and its converse [1, 4]. $\square$

Corollary 2.2 (Modal secular sum). *In the coercive branch, let $M_\delta u_k = \mu_k u_k$ with $\mu_k > 0$ and write $\beta_k = u_k^* b$. Then*

$$\Delta \geq \delta \iff \sum_{k=1}^r \frac{|\beta_k|^2}{\mu_k} \geq \delta. \quad (3)$$

The weights in (3) show why the raw cross norm $\|b\|^2$ is not the complete invariant. Coupling to a weakly coercive old-packet mode is amplified by μ_k^{-1} . Conversely, a large coupling concentrated in stiff modes may be insufficient.

For a trial $x \in \mathbb{C}^r$, define the RH-90 quadratic

$$\Phi_\delta(x) = x^* M_\delta x - 2 \operatorname{Re}(x^* b) + \delta. \quad (4)$$

Proposition 2.3 (Coercive trial-defect identity). *If $M_\delta \succ 0$ and $r_x = M_\delta x - b$, then*

$$\Phi_\delta(x) = \delta - b^* M_\delta^{-1} b + (x - M_\delta^{-1} b)^* M_\delta (x - M_\delta^{-1} b) \quad (5)$$

$$= \delta - b^* M_\delta^{-1} b + r_x^* M_\delta^{-1} r_x. \quad (6)$$

Consequently, for $\delta > 0$ and relative secular surplus

$$\kappa_\delta = \frac{b^* M_\delta^{-1} b}{\delta} - 1,$$

the trial sign is certified whenever $r_x^* M_\delta^{-1} r_x \leq \kappa_\delta \delta$.

Proof. Expand the square in (5). Substituting $x - M_\delta^{-1} b = M_\delta^{-1} r_x$ gives (6). $\square$

This identity separates three effects. The exact secular surplus is a property of the compressed dynamics, the residual defect measures the trial solve, and the absolute scale is δ . A Schur margin of order 10^{-16} may therefore coexist with a meaningful relative surplus when the requested gain is itself of order 10^{-13} . Ill conditioning can make the trial difficult to discover, but once (4) is evaluated with a strict sign, no inverse occurs in the final Rayleigh proof. This distinction is standard in validated linear algebra [3, 6].

3 Variable one-step budgets and block bootstrap

For an isometric packet V , write

$$\mathcal{E}_G(V) = \operatorname{tr} G - \operatorname{tr}(V^* G V).$$

Let $E > 0$ be the tail assigned to the previous refresh packet and let $C = \mathcal{E}_G(V)$ be its predictor tail in the new Gramian. A target factor $\rho \geq 0$ requests gain

$$\delta = C - \rho E. \quad (7)$$

Unlike RH-90, we do not require the same ρ at every update.

Theorem 3.1 (Variable-budget correction and refresh). *Let U be the leading rank- r Ritz packet in $\text{Ran}[V, q]$, and let P^+ be any rank- r refresh packet satisfying*

$$\mathcal{E}_G(P^+) \leq \mathcal{E}_G(U). \quad (8)$$

Fix $\rho \geq 0$ and define δ by (7). If $\delta \leq 0$, the old packet already gives $\mathcal{E}_G(P^+) \leq C \leq \rho E$. If $\delta > 0$ and either branch of Theorem 2.1 holds, then

$$\mathcal{E}_G(P^+) \leq \mathcal{E}_G(U) = C - \Delta \leq \rho E. \quad (9)$$

The same conclusion follows from any trial with $\Phi_\delta(x) \leq 0$.

Proof. The old coordinate packet captures $\text{tr } A$. The leading rank- r Ritz packet captures $\text{tr } H - \lambda_{\min}(H) = \text{tr } A + \Delta$, hence its tail is $C - \Delta$. The dichotomy or trial inequality gives $\Delta \geq \delta$. Insert (7) and then use (8). $\square$

The refresh condition is deliberately stated as an energy inequality. It does not require a spectral gap or principal-angle estimate. An exact leading packet satisfies it by Ky Fan's principle, but a reduced algorithm must prove it by some other means [2, 5].

Theorem 3.2 (Block contraction budget). *Let $E_n \geq 0$ be packet tails and suppose*

$$E_{n+1} \leq \rho_{n+1} E_n, \quad \rho_{n+1} \geq 0. \quad (10)$$

Fix a block length L . Assume that after a burn-in index b , every complete block satisfies

$$\prod_{s=1}^L \rho_{b+kL+s} \leq Q < 1, \quad (11)$$

$$\max_{0 \leq t \leq L} \prod_{s=1}^t \rho_{b+kL+s} \leq B \quad (12)$$

for all $k \geq 0$, with the empty product equal to one. Then

$$E_{b+kL+t} \leq BQ^k E_b, \quad 0 \leq t \leq L. \quad (13)$$

For the normalized memory Gramian with parameter $0 \leq \eta < 1$, $E_b \leq (1 - \eta)^{-1}$ and the current relative snapshot residual obeys

$$\frac{\|X_{b+kL+t}(I - P_{b+kL+t})\|_2}{\|X_{b+kL+t}\|_2} \leq \sqrt{\frac{BQ^k}{1 - \eta}}. \quad (14)$$

Proof. Multiply (10) over k complete blocks and then over the first t factors of the next block. This gives (13). Every normalized snapshot Gramian has trace one, so the memory trace is at most $(1 - \eta)^{-1}$. The current snapshot is a nonnegative weight-one summand of the memory residual, proving (14). $\square$

Corollary 3.3 (Four-step sub-quarter budget). *If $L = 4$, $Q \leq 0.24^4$, and $B = 1$, then complete block endpoints have the same average exponential rate as pointwise 0.24 contraction. At $\eta = 1/512$, the crude bootstrap reaches relative tolerances $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}$ after respectively 8, 16, 20, 28, 36, 40 updates.*

The block endpoint count is rounded to a multiple of four. It is not a claim that the frozen four-step window repeats. Establishing repeated blocks with uniform Q is the new analytic gate.

4 Validated four-step audit

4.1 Frozen models, clocks, and windows

We use the same five directional Gaussian models as RH-77–RH-91, with

$$\sigma \in \{0.16, 0.08, 0.04, 0.02, 0.01\}, \quad \eta = 1/512,$$

fine dimensions 32, 64, 128, 256, 512, and clock ranks 4, 5, 6, 6, 7. If M_σ is the archived postblock horizon, the four-step window ends at

$$t_\sigma = \max \left\{ 4, \left\lceil \frac{2M_\sigma}{3} \right\rceil \right\}.$$

Thus the audited windows are 1–4, 3–6, 8–11, 14–17, and 19–22 as the noise decreases.

At each update we form the previous leading packet, select the leading left singular direction of the new cross block, solve the $(r+1)$ -dimensional Ritz problem, and then recompute a leading refresh packet. The rank-one direction is canonical for maximal one-direction cross coupling, but the theorem itself uses only the resulting compression.

4.2 Frozen rational budgets

A binary64 pilot was used only to choose upward-rounded thousandths. The resulting integers were then frozen in the source file before the 384-bit audit. Each target is therefore the exact rational number $\rho_j = n_j/1000$, and every block product is an exact rational product. This target-selection/validation separation is important when the worst block has only a few percent of product slack.

The factors need not be below 0.24. Seven are larger, with maximum 0.753. All forty are below one, so the audited within-block prefix constant is $B = 1$.

4.3 Three independent finite checks

The audit uses three checks at every update.

First, the small compressed matrix and trial vector are lifted exactly from their binary representations and $\Phi_\delta(x)$ is evaluated in Arb ball arithmetic. Thirty-nine trials use the coercive solve; one uses an explicit negative direction of M_δ . The sign, rather than a floating inverse, is the certificate.

Second, ambient residuals are evaluated by the metric-corrected identity

$$\mathcal{E}_G(W) = \operatorname{tr} G - 2 \operatorname{tr}(W^*GW) + \operatorname{tr}((W^*W)(W^*GW)). \quad (15)$$

This is the exact energy of $(I - WW^*)$ for the lifted binary basis and remains valid when W^*W differs slightly from the identity. We directly verify the target contraction and the refresh dominance inequality. The largest basis orthogonality defect in the archive is below 2.24×10^{-15} .

Third, whenever the direct corrected contraction exceeds 0.24, we form the pointwise threshold matrix in (2). Strict positivity of every leading principal minor certifies positive definiteness by Sylvester’s criterion. Hence the named compressed correction cannot meet the pointwise target. We archive both determinant lower bounds and the associated positive pivots.

This three-layer design avoids conflating an almost orthonormal floating basis with an exact isometry. The Schur sign, ambient contraction, and refresh dominance are each checked in their own frozen model.

σ	side	rank	times	rational factors	budget mean	actual upper	slow steps
0.16	L	4	1–4	.010/.006/.004/.003	.005180	.004514	0
0.16	R	4	1–4	.017/.007/.005/.003	.006500	.006109	0
0.08	L	5	3–6	.014/.007/.009/.022	.011802	.010915	0
0.08	R	5	3–6	.008/.048/.006/.009	.012000	.011628	0
0.04	L	6	8–11	.219/.091/.064/.031	.079297	.078268	0
0.04	R	6	8–11	.050/.099/.142/.059	.080248	.079692	0
0.02	L	6	14–17	.753/.142/.126/.239	.238212	.237514	1
0.02	R	6	14–17	.145/.277/.360/.120	.204095	.203851	2
0.01	L	7	19–22	.336/.251/.169/.190	.228120	.227778	2
0.01	R	7	19–22	.391/.251/.123/.156	.208315	.207707	2

Table 1: Four-step budgets. “Budget mean” is the exact rational product to the power $1/4$; “actual upper” is the interval upper bound for the product of direct corrected contractions. A slow step is one whose corrected contraction is rigorously above 0.24.

5 Results: seven slow steps, ten contracting blocks

The hardest channel is the left channel at $\sigma = 0.02$. Its rational budget product is

$$0.753 \cdot 0.142 \cdot 0.126 \cdot 0.239 = 0.003219969564 = 0.97052516 \cdot 0.24^4.$$

The direct interval product is at most 0.003182397980, with geometric mean 0.23751368. The first step is much slower than 0.24, but the next three steps compensate.

validated quantity	worst certified value
strict Schur signs	40/40
refresh dominance inequalities	40/40
pointwise 0.24 positive-definite obstructions	7
largest four-step budget mean	0.2382116195
largest direct four-step mean upper	0.2375136767
smallest coercive relative-surplus lower	1.1169×10^{-4}
smallest absolute negative Schur margin	7.3996×10^{-18}
largest trial-system condition number	2.623×10^{13}
largest refresh/corrected tail ratio	0.99996914
smallest obstructed one-step contraction	0.25042832
smallest positive Sylvester pivot	2.8254×10^{-14}

Table 2: Global 384-bit validation summary. The relative-surplus lower bound is obtained directly from the negative trial form and requested gain.

The seven pointwise obstructions occur at

$$(0.02, L, 14), (0.02, R, 15), (0.02, R, 16),$$

and

$$(0.01, L, 19), (0.01, L, 20), (0.01, R, 19), (0.01, R, 20).$$

For each, the 0.24 threshold matrix is strictly positive definite. Thus there is no trial vector hidden by a poor solve: the exact compressed gain is below the requested pointwise gain. This is a finite no-go result for the named target, window, rank, and one-direction enrichment. It is not a no-go theorem for a different complement dimension, a different rank staircase, or an eventual later-time regime.

The refresh check is numerically close in one coarse update: the largest refresh-to-corrected ratio is 0.99996914. It nevertheless remains strictly below one at 384 bits. At the fine scales the same ratio is generally farther from one, but no uniform reduced-refresh theorem follows from these values.

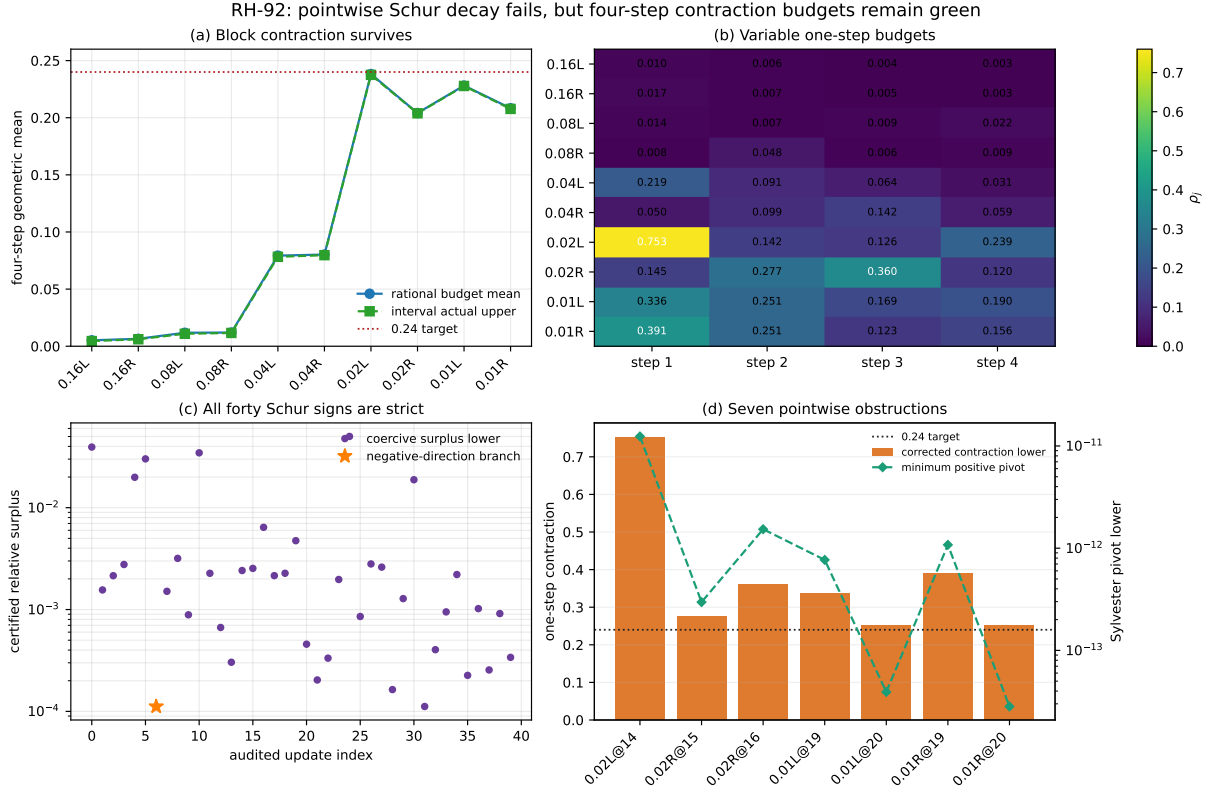


Figure 1: The block budget versus the pointwise barrier. Panel (a) shows that all ten four-step geometric means remain below 0.24. Panel (b) displays the exact rational factors. Panel (c) gives certified relative Schur surplus lower bounds, with the single negative-direction branch marked separately. Panel (d) shows the seven pointwise obstructions and their positive Sylvester pivots.

6 Route consequence and theorem boundary

Let L denote the inherited full-block corridor, R the polylogarithmic reduced packet refresh/future gate, and O the finite-prefix, normalization, and observability bridge. RH-91 used S_1 for a pointwise late Schur contraction law. The present theorem shows that the sufficient operator gate may instead be weakened to

$$S_{\text{blk}} : \prod_{s=1}^{L_0} \rho_{j+s} \leq Q < 1 \quad \text{with a uniform finite prefix factor,}$$

for some fixed block length L_0 . The revised sufficient Stage-A formula is therefore

$$A_1 = L \vee (S_{\text{blk}} \wedge R \wedge O). \quad (16)$$

The frozen evidence suggests $L_0 = 4$ and $Q = 0.24^4$, but does not prove that choice uniformly.

This adjustment helps in two ways. First, it removes seven artificial one-step failures from the preferred corridor. Second, the exact dichotomy reduces each coercive update to the resolvent-weighted scalar $b^* M_\delta^{-1} b / \delta$. An all-level proof can now aim at an averaged lower bound for secular surplus over four updates rather than a uniform one-update lower bound.

It also sharpens what has *not* been solved:

- Only one four-step window per frozen channel is certified. Repetition after an analytic burn-in remains open.

- The rational factors are data-driven. No formula bounds their product from the continuum dynamics.
- The audit refreshes with a leading ambient packet. Replacing this by a rank- $O(\log(1/\sigma))$ reduced future is exactly gate R .
- The pointwise obstruction applies only to the named rank-one enrichment. It does not exclude a two-direction corrector or rank jump.
- Gates O and the RH-81 moving-cloud gates P, C, U are unchanged [10, 12].

No canonical scattering completion, self-adjoint generator, intrinsic $T \log T$ counting law, prime-power trace formula, completed-zeta identity, Hilbert–Polya operator, zeta-zero identification, or proof of the Riemann Hypothesis is obtained here.

7 Reproducibility

The archive contains the full and smoke audits, exact rational budget manifest embedded in source, 384-bit interval records for all forty updates, all Sylvester minors and pivots for the seven obstructions, the generated figure, unit tests for the analytic formulas, and SHA-256 manifests for local sources, external inputs, and publication artifacts. The principal commands are listed in the accompanying README. All model matrices and trial vectors are lifted from their binary representations; no decimal reconstruction is used in a sign test.

References

- [1] Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.
- [2] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, fourth edition, 2013.
- [3] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. SIAM, second edition, 2002.
- [4] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, second edition, 2013.
- [5] Beresford N. Parlett. *The Symmetric Eigenvalue Problem*. SIAM, 1998.
- [6] Siegfried M. Rump. Verification methods: Rigorous results using floating-point arithmetic. *Acta Numerica*, 19:287–449, 2010.
- [7] Liang Wang. Trace-normalized late-memory packets. 2026. RH-86 manuscript.
- [8] Liang Wang. From half-log rank clocks to schur packet bootstrap. 2026. RH-91 manuscript.
- [9] Liang Wang. Rank-one complement ritz correction for dynamic packet energy. 2026. RH-89 manuscript.
- [10] Liang Wang. From validated assembly to moving-cloud renormalization. 2026. RH-81 manuscript.
- [11] Liang Wang. Schur–secular sub-quarter contraction certificates. 2026. RH-90 manuscript.
- [12] Liang Wang. Two corridors to stage a1. 2026. RH-78 manuscript.